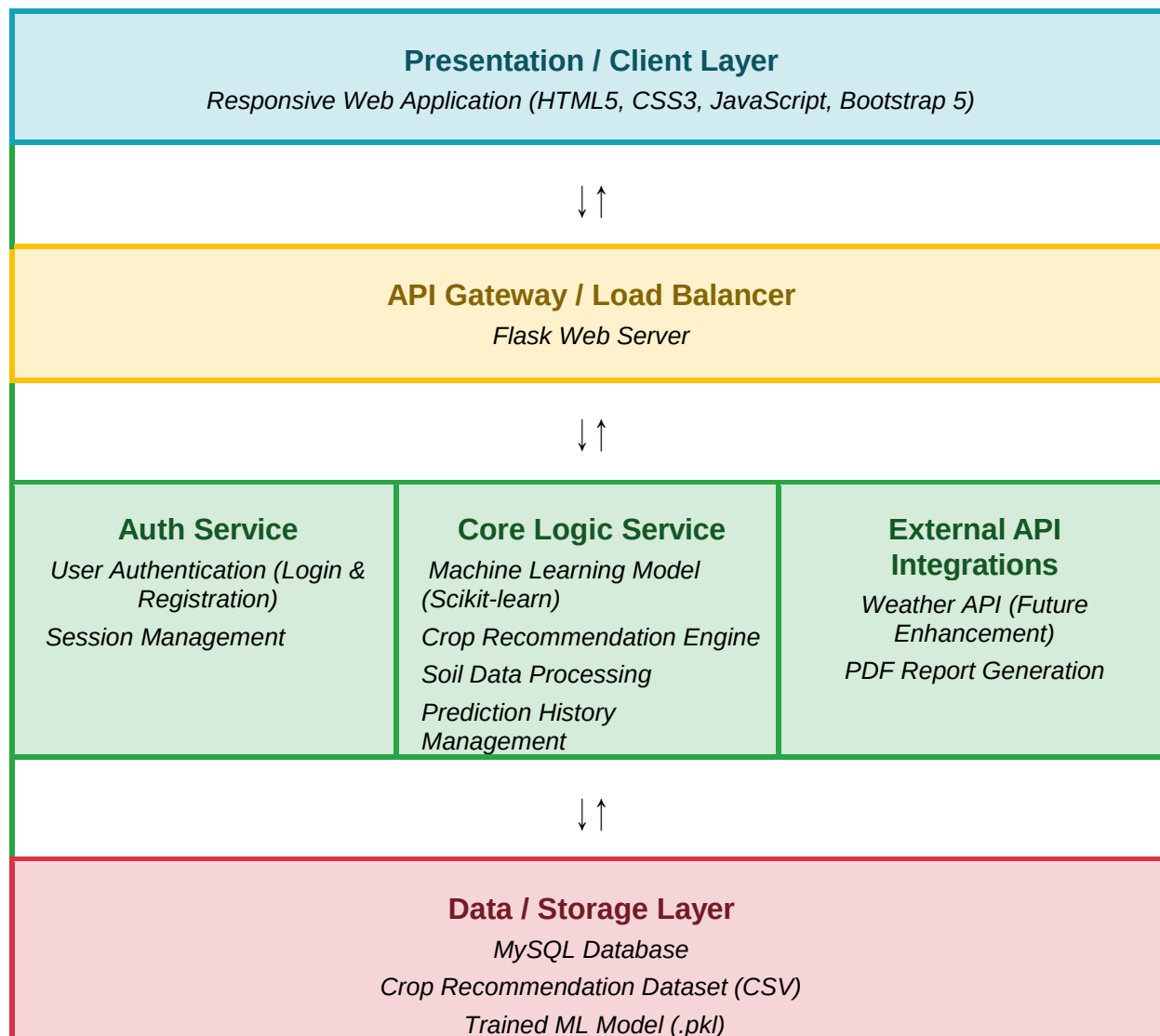


Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides a responsive web interface for user registration, crop prediction, and viewing results.</i>	<i>HTML5, CSS3, JavaScript, Bootstrap 5</i>
<i>API Gateway</i>	<i>Handles user requests, routes data between the frontend and backend, and manages application logic.</i>	<i>Flask</i>
<i>Microservices</i>	<i>Processes soil parameters, executes the Machine Learning model, generates crop recommendations, and stores prediction history.</i>	<i>Python, Scikit-learn, Pandas, NumPy</i>
<i>Database</i>	<i>Stores user information, prediction history, crop data, and machine learning resources.</i>	<i>MySQL, CSV Dataset, Joblib (.pkl Model)</i>